# Computational Analysis of Ligament Force Sensitivity to Attachment Points during Squatting

Viktor Klassen[1], JH Muller[1], L Labey[2]

**Abstract** Ligaments are bands of strong, elastic tissue that give support and limit the joint's movement. By being connected to one bone and to an adjacent bone ligaments make it possible to perform various movements. Despite its toughness ligaments can be damaged caused by sport injuries or diseases. To restore joint stability, trauma is often treated by replacing the equivalent ligament. For this the torn or insured ligament is removed before the tissue graft is inserted at its new position.
The aim of this study was to analyse, using computational modelling techniques, whether the tension in the ligaments surrounding the knee joint differ when the ligament insertion site is shifted in specific direction. For this, the sensitivity of different implant positions was compared with one another and areas with high, middle or low sensitivity zoned. The methods for defining the tensile force, as well as the different stiffness values for the ligaments were derived inter alia from Wismans et al. and Blankevoort et al.
Obtained results show that different insertion sites have indeed an effect on the strain intensity. In addition one can see that the tensions vary partly tremendously when the ligaments are shifted into another direction. The study can be of interest for clinicians when performing ligament reconstruction.

**Index Terms:** Computational modelling techniques, Reconstruction, Tensile Force, Tension, Sensitivity and Implant Position.

[1]Viktor Klassen, Dr. JH Muller
Department of Mechanical and Mechatronic Engineering, Biomedical Engineering, University of Stellenbosch, South Africa
[2]Luc Labey, KU Leuven, Mechanical Engineering Technology TC, Belgium
Viktor Klassen
Department of Mechanical and Mechatronic Engineering, Biomedical Engineering, University of Stellenbosch
Stellenbosch, 7600, South Africa
e-mail: Klassen_viktor@gmx.net

## I. Introduction

The knee is the largest and most complex joint in the human body. It consists of many muscles, ligaments and bone structures[1]. The knees provide the required support for the body during different movements such as standing, walking, running, jumping, etc. For this, the knee needs to be flexible and stable at the same time. The anatomical structure of the knee makes this possible but also leads to the knee being susceptible to injuries. Damage can be caused by impact loading, which is often the case in sports injuries[2]. In some cases, normal healing is insufficient and surgical repair of soft tissues is needed.
Over the last few years, several studies have been performed to investigate whether and how a chosen position during ligament reconstruction affects the clinical outcome. According to Kupczik et al. the femoral anatomical origin point of the medial patellofemoral ligament (MPFL) is important when ligament reconstruction is taking place. He showed that a non-anatomical position can lead to ligament overload, medial subluxation of the patella or excessive tensioning of the graft. It is still controversial as to which position is truly the best choice for placing the graft[3]. The same applies to the other ligaments of the knee joint. C.F Jepsen et al. evaluated whether a change in the femoral graft insertion site between the 1-o'clock (high) and 2-o'clock (low) positions could change the laxity of the anterior cruciate ligament (ACL). The study contained a prospective, randomized, double blind investigation. Sixty patients were divided into a low tunnel and high tunnel position group. Published conclusion showed that it is possible to improve the clinical result in 1-bundle ACL reconstruction by lowering the femoral tunnel position. Despite the fact that the sagittal laxity did not change detectable, a follow-up survey with the patients, however, showed significantly that the lower graft position was favoured[4].

We hypothesise that ligaments will be susceptible to changes in implant position. The aim is to determine whether the ligament response (i.e. tension) will be greater/more sensitive if the attachments are shifted in specific directions. This may enable the definition of safe zones in which a ligament's response will remain consistent, or directions to increase ligament function.

Computational models can and have been used to evaluate knee biomechanics[5,6]. Prior computational studies have taken place that investigate the effect of repositioning attachment points of soft tissues. However, limitations persist. O. Kwon et al. used a computational model to investigate the influence of the ligament bundle diameter and attachment point on kinematic behaviour in double bundle ACL. His research results shows that the knee kinematics after the double bundle ACL reconstruction was dependent on the attachment point and less dependent on the diameter size. Based on this, it was recommended to focus on the attachment point rather than the diameter.

## II. Materials and Methods

The data utilized was derived from a prior in vitro knee rig study on cadaveric knees. The description of the experiment, its methodology and kinematic analysis can be found in Victor et al., 2009[7] and Delport et al.. For our application we used one dataset with the above-mentioned content. Although the dataset contained the three-dimensional surface model, modifications were needed to be done due to its high resolution. For this, the segments were imported into a medical image processing software (Mimics 11.02, Materialise, Leuven, Belgium). With Mimics and its powerful adaptive filtering tool a significant reduction of the file size took place[8]. After the procedure, the modified model was imported into a multibody dynamic software (ADAMS, MSC Software Corporation, Santa Ana, CA)[9] with which the squat movement of the knee rig study was simulated. In addition ADAMS found its application in creating the ligaments. MATLAB (MathWorks Inc., Natick, MA, USA) was used for the calculations of the tensile force, the strain and the sensitivity analysis.

**Table 1 – Investigated ligaments with its ligament bundles. Information in source was used to calculate stiffness parameter.**

| Ligament | Ligament Bundles | Source |
|---|---|---|
| Posterior Cruciate Ligament (PCL) | Posteromedial PCL (PMPCL) | M. Gupte, 2006, |
| | Anterolateral PCL (ALPCL) | J. Gali et al., 2013 |
| Anterior Cruciate Ligament (ACL) | Posterolateral ACL (PLACL) | W. Petersen et al., 2015, |
| | Anteromedial ACL (AMACL) | E. Bicer et al., 2010 |
| Lateral Collateral Ligament (LCL) | LCL | J. Brinkman et al., 2005 |
| Medial Collateral Ligament (MCL) | Deep Medial Collateral Ligament (dMCL) | M. Laprade et al., 2015, F. Liu et al., 2010 |
| Patella Tendon (PT) | Medial PT (MPT) | |
| | Central PT (CPT) | |
| | Lateral PT (LPT) | |
| Medial Patellofemoral Ligament (MPFL) | MPFL20 | J. Victor et al., 2009, |
| | MPFL40 | R. Steensen et al. 2004 |
| | MPFL60 | |

In this study, however, cadaveric specific data that was measured from an in-vitro set up was used to create a three-dimensional knee model. The model contained optical reflective reference markers, anatomical landmarks and axes. Moreover, the movement of the squat motion was given.

Basically, the model consisted of the distal part of the femur, the proximal part of the tibia and the patella. Once the model was imported into ADAMS, twelve markers were created according to the optical

reference markers used for the Delport et al., 2014 study. The motion of each marker was defined according to data obtained from a three-dimensional motion capture system (Vicon Motion Systems, Los Angeles, California) which recorded the trajectory of each marker during the squat movement. The recorded trajectories were applied to the markers in our computational model.

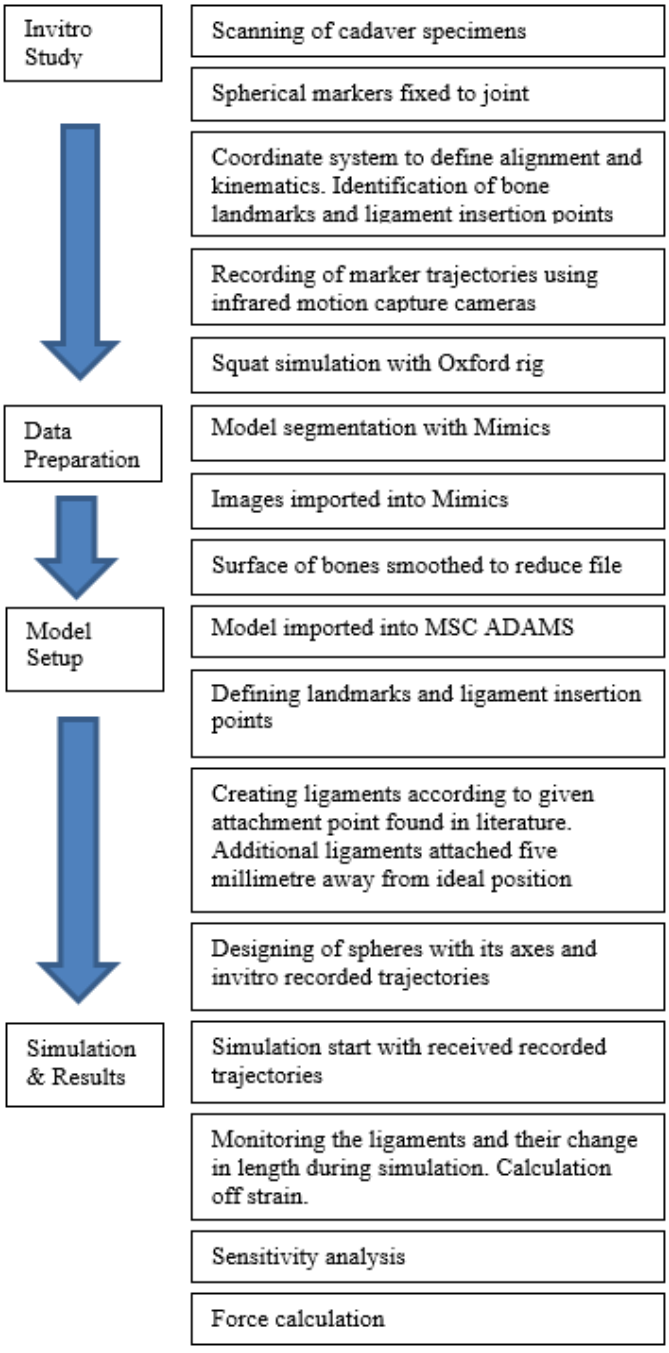



**Figure 1 – Flow Chart**

Since the data that was received did not contain information for all ligaments and their origin and insertion points, additional literature in combination with the model specific anatomical landmarks were used to define these points (Table 1). These points will be further mentioned as most ideal positions for the ligaments attachments.

The distance between created points indicated the ligament length and was measured with a Point-to-Point measuring tool in ADAMS. In total, six ligaments, made up of several ligament bundles, were created. Origin and insertion points starting from a specific anatomical landmark were not always available. Estimations were made for those points. However, for some ligaments, the description that was found gave a good indication. Created ligaments only have two connection points, which are located between the tibia- femur, tibia- patella or femur-patella. Due to the width of the medial patellofemoral ligament (MPFL) and patella tendon (PT) the ligaments were divided into 3 ligaments bundles (Table1). The relevant connection points for the MPFL were found in J. Victor et al. 2009 for the patella and in Robert N. Steensen et al., 2004 for the femur.

**Table 2 – Calculated average stiffness parameter**

| Ligament | Average Stiffness Parameter |
|---|---|
| PMPCL | 8300.08[10,11,12,13,14] |
| ALPCL | 13830.39[10,11,12,13,14] |
| PLACL, AMACL | 7522.10[10,11,12,13,14,15] |
| LCL | 7434.13[10,11,12,13,14,16] |
| dMCL | 4206.99[11,12,13,14,17] |
| MPT, CPT, LPT | 29096.35[13,18,19,20,21] |
| MPFL20, MPFL40, MPFL60 | 285.70[13,22,23,24,25] |

Within the scope of this study the ligaments were assumed to be elastic, which means that the force during the knee locomotion is a function of its length. Each ligament is defined using the force-displacement curve, that was first introduced by Wismans et al.[27] and Blankevoort et al.[14]. The ligament modelling terminology by Blankevoort et al.[14] is used. The function is assumed to be non-linear for low strains and linear for strains higher than $2\varepsilon_l$ ($\varepsilon_l = 0.03$)[14,15]:

$$f = \begin{cases} \frac{1}{4}k\varepsilon^2/\varepsilon_l, & 0 \le \varepsilon \le 2\varepsilon_l, \\ k(\varepsilon - \varepsilon_l), & \varepsilon > 2\varepsilon_l, \\ 0, & \varepsilon < 0, \end{cases}$$

Where $f$ is the tensile force of the ligament, k is the ligament stiffness, and ε the strain in the ligament, based on the instantaneous ligament length L and its zero-load length $L_0$:

$$\varepsilon = (L - L_0)/L_0.$$

The zero-load length was defined as the distance between the respective attachment points, obtained from the joint in full extension.
Based on various stiffness parameters, which can be found in numerous literature sources, an average of those was taken for each ligament[23,21,12,13]. Often articles did not state the stiffness parameter, but did report on the ligament's cross-sectional area (A) and its elasticity modulus (E), which were used to calculate the stiffness parameter in the equation below (Table 2).

$$k = AE\ (Newton)$$

Four additional markers were placed five millimetres away from the ideal marker position in the anterior, posterior, medial, lateral, proximal or distal directions, respectively (Figure 2).

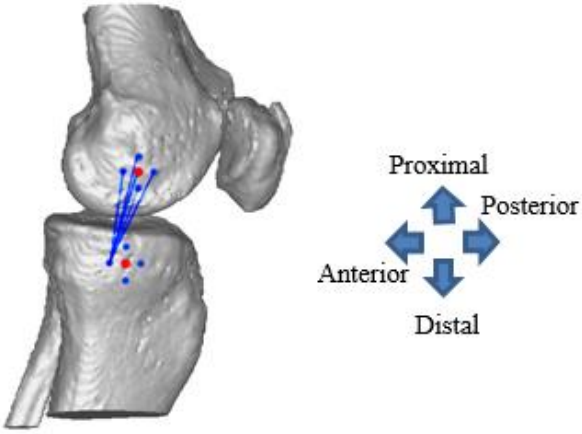



**Figure 2 - Knee joint with its directions and ligament attachment points. In red: the previously defined ideal position for respective ligament. In blue: Additional markers five millimetre away. In total there are 25 ligament connection combinations.**

All markers were connected with each other and the distances between them were monitored during the squat simulation. In total there are 25 ligament combinations for each ligament. The MPFL is the only exception with 20 connections for each bundle. Reason for that is the limited space on the patella in anterior direction, which made it impossible to create a marker. Finally, a sensitivity analysis with MATLAB considering all combinations of one ligament took place.

$$\hat{S} = (max[\Delta F_Lig\ ]/F(LigIdeal, max)$$

For this equation, the maximum force of one of the four attachment combinations was subtracted from the force of the ideally attached one and further divided by the maximum force of the ideal ligament.

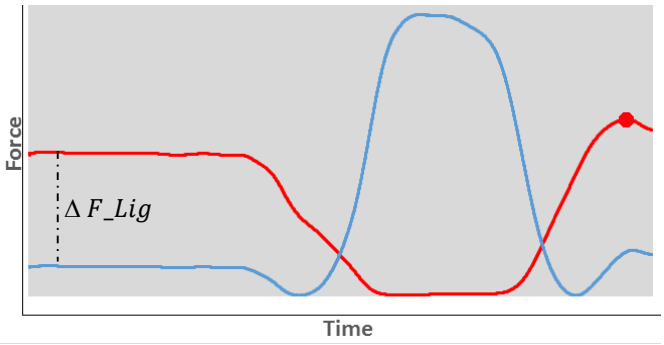



**Figure 3 – Tension within the ligament during squatting. Red point = maximal tension at ideal position (see: LigIdeal,max).**

Obtained sensitivity values were collected and zoned according to the ligament and its particular direction. Ligament connections that showed great response in their sensitivity were monitored for their tensions during squatting and compared to results from other studies. This, however, aimed to validate the results.

## III. Results

In the form of a heat map it is easy to see the attachment combination with the highest sensitivity (Table 3). For the LCL, for example, the highest sensitivity for the femur and tibia attachment is found to be posterior from the ideal central position with a result of 1.32. The three ligaments with the highest sensitivities are: MPFL (6.41), PLACL (6.30) and PMPCL (2.53). For the MPFL60 the mentioned sensitivity is found for the patella in proximal and for the femur in distal direction. The directions for the PLACL are for the femur in posterior and for the tibia in lateral direction.
Not as high as the PLACL and MPFL60, the PMPCL has the third highest result in its sensitivity with 2.53 in the distal direction of the femur and the medial direction for the tibia. Obtained results do also show

middle and low range sensitivity values. The lowest sensitivity value was found for the MPFL40 ligament. With a negative value of -0.95 the ligament was attached on the femur in proximal and on the patella in the most ideal central position. It is noticeable that the results for each ligament bundle are clearly structured. Low, middle or high sensitivity values are assigned to specific directions. For instance one can see that for the dMCL the range for high sensitivities is found for the femur in anterior and distal direction, while low sensitivities are seen in posterior and proximal direction. Another example is the MPFL20 ligament that shows high range values for the femur in distal and posterior direction, while low range values are only found in proximal direction.

**Table 3 – Sensitivity analysis of all 12 ligaments and their 25 different combinations (heat map)**

| Tibia \ Femur | PMPCL | Anterior | Distal | Posterior | Proximal |
|---|---|---|---|---|---|
| PMPCL | 0,00 | 0,41 | 2,03 | 0,19 | -0,14 |
| Anterior | 0,11 | 0,53 | 2,42 | 0,29 | -0,02 |
| Lateral | -0,18 | 0,15 | 1,50 | -0,05 | -0,22 |
| Medial | 0,16 | 0,62 | 2,53 | 0,53 | -0,05 |
| Posterior | -0,10 | 0,27 | 1,70 | 0,11 | -0,23 |

| Tibia \ Femur | PLACL | Anterior | Distal | Posterior | Proximal |
|---|---|---|---|---|---|
| PLACL | 0,00 | 5,92 | 1,72 | 4,65 | -0,10 |
| Anterior | 0,21 | 5,19 | 0,95 | 3,93 | -0,52 |
| Lateral | 0,38 | 6,03 | 2,39 | 6,30 | -0,34 |
| Medial | -0,61 | 4,75 | 1,24 | 2,72 | -0,09 |
| Posterior | -0,59 | 5,17 | 4,91 | 5,23 | 0,16 |

| Tibia \ Femur | LCL | Anterior | Distal | Posterior | Proximal |
|---|---|---|---|---|---|
| LCL | 0,00 | -0,53 | -0,68 | 0,93 | 0,78 |
| Anterior | -0,43 | -0,43 | -0,81 | 0,55 | 0,26 |
| Distal | -0,17 | -0,53 | -0,79 | 0,73 | 0,57 |
| Posterior | 0,42 | -0,27 | -0,35 | 1,32 | 1,29 |
| Proximal | 0,26 | -0,46 | -0,49 | 1,26 | 1,09 |

| Tibia \ Femur | dMCL | Anterior | Distal | Posterior | Proximal |
|---|---|---|---|---|---|
| dMCL | 0,00 | 0,82 | 0,94 | -0,69 | -0,57 |
| Anterior | -0,08 | 0,79 | 0,71 | -0,82 | -0,56 |
| Distal | -0,18 | 0,51 | 0,55 | -0,76 | -0,64 |
| Posterior | 0,03 | 0,77 | 1,05 | -0,56 | -0,60 |
| Proximal | 0,25 | 1,26 | 1,54 | -0,59 | -0,47 |

| Tibia \ Femur | AMACL | Anterior | Distal | Posterior | Proximal |
|---|---|---|---|---|---|
| AMACL | 0,00 | -0,76 | 0,16 | 0,86 | -0,16 |
| Anterior | -0,19 | -0,61 | -0,03 | 0,60 | -0,34 |
| Lateral | 0,17 | -0,72 | 0,31 | 1,13 | 0,02 |
| Medial | -0,15 | -0,76 | -0,05 | 0,60 | -0,28 |
| Posterior | 0,26 | -0,68 | 0,41 | 1,19 | 0,07 |

| Tibia \ Femur | ALPCL | Anterior | Distal | Posterior | Proximal |
|---|---|---|---|---|---|
| ALPCL | 0,00 | 0,61 | 1,35 | -0,58 | -0,32 |
| Anterior | 0,23 | 0,96 | 1,92 | -0,54 | -0,15 |
| Lateral | -0,19 | 0,42 | 0,98 | -0,60 | -0,43 |
| Medial | 0,03 | 0,55 | 1,31 | -0,50 | -0,31 |
| Posterior | -0,18 | 0,32 | 0,88 | -0,49 | -0,27 |

| Tibia \ Patella | MPT | Distal | Lateral | Medial | Proximal |
|---|---|---|---|---|---|
| MPT | 0,00 | -0,10 | 1,14 | -0,31 | 0,05 |
| Distal | 0,39 | 0,31 | 1,51 | -0,46 | 0,43 |
| Lateral | -0,37 | -0,26 | 0,57 | 0,04 | -0,32 |
| Medial | 0,18 | 0,18 | 1,08 | -0,07 | 0,15 |
| Proximal | -0,21 | 0,04 | 0,58 | 0,43 | -0,27 |

| Tibia \ Patella | CPT | Distal | Lateral | Medial | Proximal |
|---|---|---|---|---|---|
| CPT | 0,00 | -0,01 | 0,59 | -0,55 | -0,01 |
| Distal | 0,37 | 0,40 | 0,95 | -0,18 | 0,32 |
| Lateral | -0,48 | -0,51 | -0,03 | -0,54 | -0,45 |
| Medial | 0,39 | 0,49 | 0,99 | -0,19 | 0,31 |
| Proximal | -0,31 | -0,36 | 0,31 | -0,58 | -0,27 |

| Tibia \ Patella | LPT | Distal | Lateral | Medial | Proximal |
|---|---|---|---|---|---|
| LPT | 0,00 | 0,06 | 0,57 | -0,37 | 0,01 |
| Distal | 0,54 | 0,62 | 1,09 | -0,02 | 0,49 |
| Lateral | -0,29 | -0,25 | -0,02 | -0,22 | -0,27 |
| Medial | 0,77 | 0,95 | 1,41 | 0,16 | 0,68 |
| Proximal | -0,19 | -0,12 | 0,28 | -0,34 | -0,17 |

| Femur \ Patella | MPFL20 | Distal | Posterior | Proximal |
|---|---|---|---|---|
| MPFL20 | 0,00 | -0,11 | 0,16 | -0,06 |
| Anterior | -0,16 | -0,28 | 0,03 | -0,21 |
| Distal | 0,99 | 0,96 | 1,30 | 0,81 |
| Posterior | 0,43 | 0,41 | 0,65 | 0,30 |
| Proximal | -0,80 | -0,89 | -0,73 | -0,78 |

| Femur \ Patella | MPFL40 | Distal | Posterior | Proximal |
|---|---|---|---|---|
| MPFL40 | 0,00 | -0,61 | 0,12 | 0,46 |
| Anterior | -0,62 | -0,93 | -0,53 | -0,17 |
| Distal | 1,53 | 0,67 | 1,88 | 2,05 |
| Posterior | 0,84 | 0,29 | 1,09 | 1,16 |
| Proximal | -0,95 | -0,93 | -0,94 | -0,79 |

| | | Patella | | | |
|---|---|---|---|---|---|
| | | MPFL60 | Distal | Posterior | Proximal |
| Femur | MPFL60 | 0,00 | -0,65 | -0,21 | 1,84 |
| | Anterior | -0,88 | -0,28 | -0,74 | -0,77 |
| | Distal | 3,74 | 1,58 | 3,71 | 6,41 |
| | Posterior | 3,11 | 1,96 | 3,24 | 4,84 |
| | Proximal | 0,28 | 1,19 | 0,69 | -0,63 |

Referring to Figure 4 one can see the radar graphs that indicate the particular sensitivities in each direction as well as the ideal position of the ligament that is in the centre of the graph. It is to be mentioned that the scale does not compile with actual measurements of the attachment points, but only

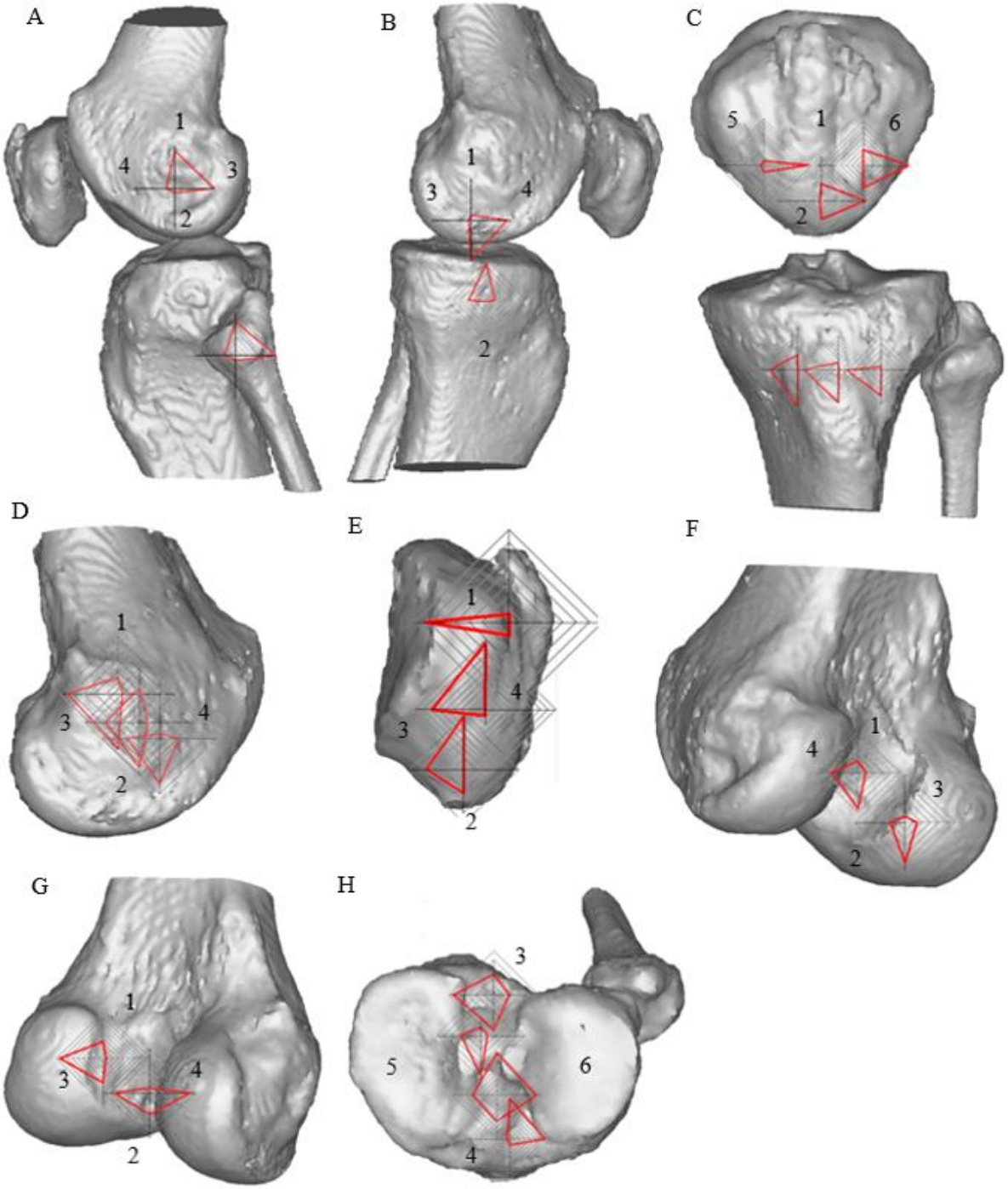



**Figure 4 – Ligaments and their sensitivity when being shifted in particular direction. 1 = proximal; 2 = distal; 3 = posterior; 4 = anterior; 5 = medial; 6 = lateral. A = LCL; B = dMCL; C = PT (from left to right MPT, CPT, LPT); D = MPFL (from left to right MPFL20, MPFL40, MPFL60); E = MPFL (Top - MPFL20, Centre – MPFL40, bottom – MPFL60); F = left – ALPCL, right - PMPCL; G = left – AMACL, right - PLACL; H = from top down – PMPCL, ALPCL, PLACL, AMACL**

serve as visualization of the different zones.
One can see that the sensitivity is high in only one or two directions for almost all ligaments. The only two exceptions are the MPFL40 and PLACL that show, in comparison to the other ligaments, a balanced graph in which the sensitivities do not differ greatly when shifted to another direction.
Monitoring the tension within the ligaments gives a good understanding of how great the effect of a shifted ligament can have when being attached into a different direction (Figure 5). The results show that the tension within the ligaments increases in accordance to a greater sensitivity. Those, however, can vary tremendously. Looking at the ALPCL it is seen that the tension can either have its maximum at 849N or 5442N during squatting, depending on the attachment point. Low tensions are specifically found in all MPFL ligaments of which the MPFL60 has the lowest tension with 0.77N to 47.1N, depending on the attachment point (Table 4 & Figure 4).

## IV. Discussion

Similar to the study of O. Kwon et al., this study also shows the importance of attachment points and its different ligament behaviour in tension.
In fact, obtained results do not only show that the ligament response is more sensitive when the attachment point is shifted, but also that the sensitivities can be zoned according to chosen direction. This, however, conforms with above stated hypothesise that ligaments are indeed susceptible to

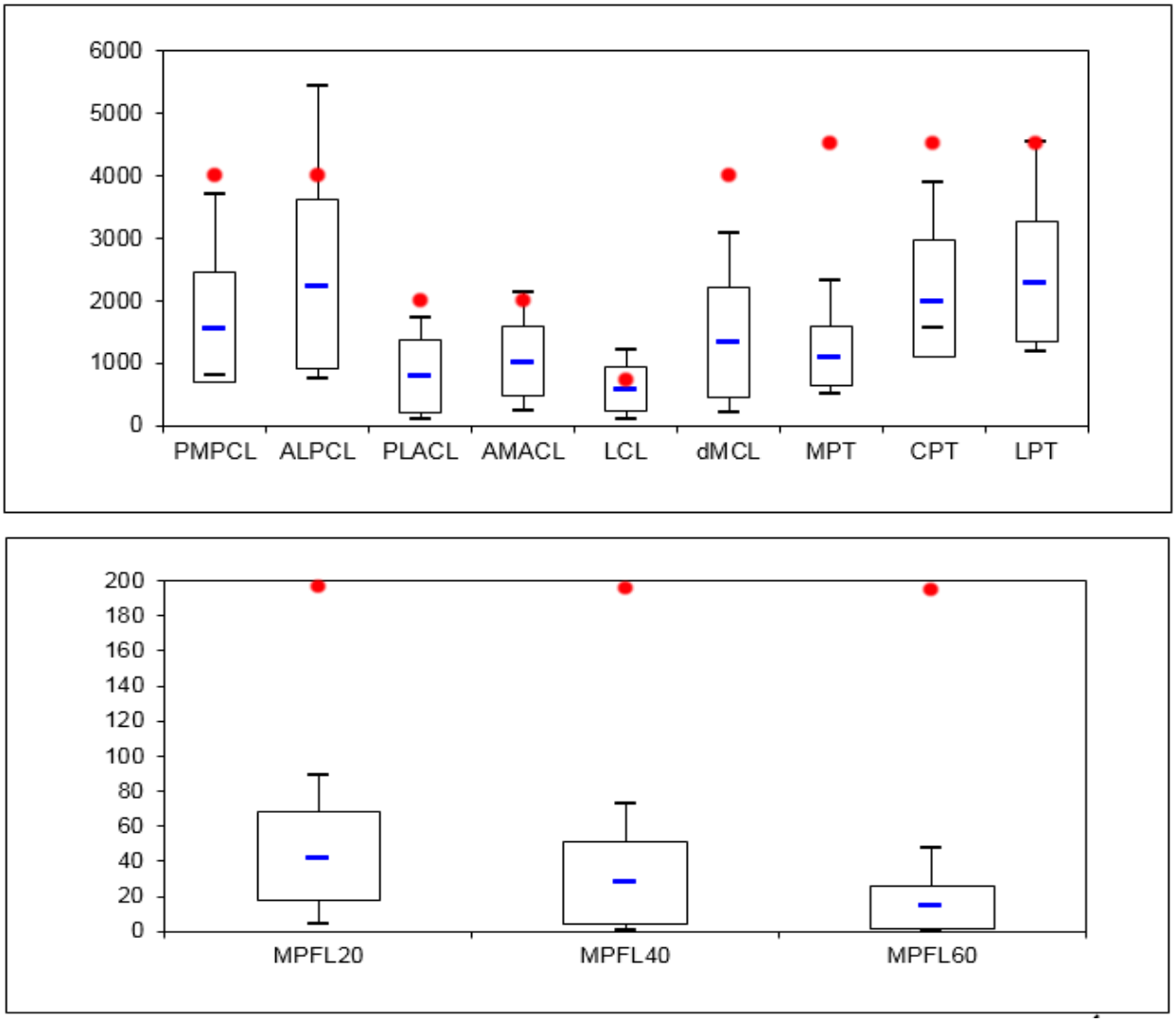



**Figure 5 – Maximum tension within the ligaments going from the connection with the lowest to the highest sensitivity. Median in Blue, Standard Deviation indicated by the box; Tension indicated by the upper and lower whiskers, Red dot = Max. load for respective ligament according to literature[30, 31, 32, 33, 34].**

changes in implant position. Although other studies have already focused on various attachment points and its effect for ligament replacement, this study, however, focussed not only on one, but six different ligaments surrounding the knee joint of one specific model using computational modelling techniques.

Focussing on table 3 the results show sensitivities that are, besides few exceptions, accurately arranged or zoned into low (Green), middle (Yellow/ Light-Orange) and high values (Dark- Orange/ Red). The sensitivity analysis was only performed for each ligament bundle and its 25 different connections, which is why a value in one bundle has a different colour to a same value in another bundle.

Practically speaking, the results could find their clinical application as follows. Say for example a patient has limitations in his musculoskeletal system caused by imbalanced ligaments. This may occur due to sports injuries or diseases that eventually lead to a varus/valgus and an indispensable surgery. If a ligament is torn a surgeon would choose an attachment point that brings back stability, but also allows the ligament to withstand higher tensions than before. According to obtained result such decision can have a great impact in the patient's mobility. Despite the fact that ligaments were only shifted five millimetres away from an anatomical ideal position, the tension varies in a large extend. Given the above mentioned example (ALPCL), also other ligaments show a great variation in tension when shifted. Figure 5 indicates besides the particular ligament tensions, also the maximum load of those. It is seen that for all obtained results, the mean value is below this load. However, for four ligaments the tension exceeds the maximal load of that particular ligament. In those cases a ligament replacement in direction with high sensitivity is not recommended. If a LCL need to be replaced an attachment point has to be chosen carefully, since the maximum load is within the calculated tension range. At this model it is recommended to choose an attachment point for the femur in distal or anterior direction, since those lay below the mean value and thus below the maximum load. This directions can be defined as safe zones. The attachment on the tibia side can then be neglected, because all values are located in the low range. Here, each direction is in a safe zone for the LCL replacement.

All tensions for the dMCL, PLACL, PMPCL, ect are located below the maximal load, which makes a replacement less critical than for ligaments like the LCL, LPT, ALPCL or AMACL for which safe zones need to be defined.

While the model could have a safe replacement in each direction for the MPFL, it cannot have this for the ALPCL. Here the replacement should not be taken place at the femur in distal direction or anterior direction, because the ligament can easily overstrain or tear.

## V. Conclusion

It is known that imbalances in ligament force can lead to subluxation, articular overload or instability[28,29]. That means that ligaments are either too lax or firm. Due to this, ligament replacements can have different purposes. A surgical decision is based on the patient specific need. Nonetheless, received results show that the direction which is chosen has a great impact on ligament tension and therewith on joint stability.

However, due to the complexity of the knee joint a general conclusion about an ideal attachment point for all ligaments cannot be made. One of the reasons is that a dataset of only one knee was used for this research. For further research it would be of interest to compare additional knee models and their sensitivities with each other. A hypothesis could be whether a save zone remains the same for the specific ligament. Therefore additional research with different knee models is needed.

The study gave a first point of view using computational modelling techniques to analyse ligaments and their behaviour during squatting. Moreover it was seen that this method can have a great impact in performing studies like these.

## VI. References


1. Lobenhoffer P. *Das Kniegelenk*. http://solutions.3maustria.at/3MContentRetrievalAPI/BlobServlet?lmd=1353673486000&locale=de_AT&assetType=MMM_Image&assetId=1319242365310&blobAttribute=ImageFile.

2. Gets W, Problems K, Causes W. What Are Knee Problems ? 2006;(March):1-5. http://www.niams.nih.gov/Health_Info/Knee_Problems/knee_problems_ff.pdf.

3. Kupczik F, Gunia Schiavon ME, Sbrissia B, de Almeida Vieira L, de Moura Bonilha T. Anchor proximal migration in the medial patellofemoral ligament reconstruction in skeletally immature patients. *Rev Bras Ortop (English Ed*. 2013;48(5):465-468. doi:10.1016/j.rboe.2013.01.002.

4. Jepsen CF, Lundberg-Jensen AK, Faunoe P. Does the Position of the Femoral Tunnel Affect the Laxity or Clinical Outcome of the Anterior Cruciate Ligament–Reconstructed Knee? A Clinical, Prospective, Randomized, Double-Blind Study. *Arthrosc J Arthrosc Relat Surg*. 2007;23(12):1326-1333. doi:10.1016/j.arthro.2007.09.010.

5. Elias JJ, Cosgarea AJ. Computational modeling: an alternative approach for investigating patellofemoral mechanics. *Sports Med Arthrosc*. 2007;15(2):89-94. doi:10.1097/JSA.0b013e31804bbe4d.

6. Powers CM, Chen YJ, Scher I, Lee TQ. The influence of patellofemoral joint contact geometry on the modeling of three dimensional patellofemoral joint forces. *J Biomech*. 2006;39(15):2783-2791. doi:10.1016/j.jbiomech.2005.10.008.

7. Victor BJ, Glabbeek F Van, Sloten J Vander, Parizel PM. An Experimental Model for Kinematic Analysis of the Knee. 2009:150-163. doi:10.2106/JBJS.I.00498.

8. Prof. Dr. J. Vander Sloten TVC. Mimics STL + Module Mimics FEA Module. 2010:1-3. www.materialise.com.

9. MSC.Software. Multibody Dynamics Analysis for Better System Simulations. 2013. http://mscsoftware.com/sites/default/files/br_adams_ltr_pt.pdf. Accessed November 6, 2015.

10. Lanzendorf, E.J., Murphy, M.C. and Mann RW. MEASUREMENT OF LIGAMENT TISSUE CROSS-SECTIONAL AREA. 1988:22-25.

11. Moglo KE, Shirazi-Adl a. On the coupling between anterior and posterior cruciate ligaments, and knee joint response under anterior femoral drawer in flexion: A finite element study. *Clin Biomech*. 2003;18(8):751-759. doi:10.1016/S0268-0033(03)00140-2.

12. Lai Y-S, Chen W-C, Huang C-H, Cheng C-K, Chan K-K, Chang T-K. The Effect of Graft Strength on Knee Laxity and Graft In-Situ Forces after Posterior Cruciate Ligament Reconstruction. *PLoS One*. 2015;10(5):e0127293. doi:10.1371/journal.pone.0127293.

13. Mesfar W, Shirazi-Adl a. Biomechanics of the knee joint in flexion under various quadriceps forces. *Knee*. 2005;12(6):424-434. doi:10.1016/j.knee.2005.03.004.

14. Blankevoort L, Kuiper JH, Huiskes R, Grootenboer HJ. Articular contact in a three-dimensional model of the knee. *J Biomech*. 1991;24(11):1019-1031. doi:10.1016/0021-9290(91)90019-J.

15. Butler DL, Kay MD, Stouffer DC. Comparison of material properties in fascicle-bone units from human patellar tendon and knee ligaments. *J Biomech*. 1986;19(6):425-432. doi:10.1016/0021-9290(86)90019-9.

16. Shetye SS, Malhotra K, Ryan SD, Puttlitz CM. Determination of mechanical properties of canine carpal ligaments. *Am J Vet Res*. 2009;70(8):1026-1030. doi:10.2460/ajvr.70.8.1026.

17. Woo SLY, Abramowitch SD, Kilger R, Liang R. Biomechanics of knee ligaments: Injury, healing, and repair. *J Biomech*. 2006;39(1):1-20. doi:10.1016/j.jbiomech.2004.10.025.

18. O'Brien TD, Reeves ND, Baltzopoulos V, Jones D a., Maganaris CN. Mechanical properties of the patellar tendon in adults and children. *J Biomech*. 2010;43(6):1190-1195. doi:10.1016/j.jbiomech.2009.11.028.

19. Couppé C, Hansen P, Kongsgaard M, et al. Mechanical properties and collagen cross-linking of the patellar tendon in old and young men. *J Appl Physiol*. 2009;107(3):880-886. doi:10.1152/japplphysiol.00291.2009.

20. Helland C, Bojsen-Møller J, Raastad T, et al. Mechanical properties of the patellar tendon in elite volleyball players with and without patellar tendinopathy. *Br J Sports Med*. 2013;47(13):862-868. doi:10.1136/bjsports-2013-092275.

21. Matschke V, Jones JG, Lemmey AB, Maddison PJ, Thom JM. Patellar Tendon Properties and Lower Limb Function in Rheumatoid Arthritis and Ankylosing Spondylitis versus Healthy Controls: A Cross-Sectional Study. *Sci World J*. 2013;2013:1-8. doi:10.1155/2013/514743.

22. Menon PG, Muller JH. Characterization of a Novel Imaging-Based Metric of Patellofemoral Separation Using Computational. 2014:188-203.

23. Ligaments P. A COMPARISON OF THE MECHANICAL AND DIMENSIONAL PROPERTIES OF THE HUMAN MEDIAL AND LATERAL 46th Annual Meeting , Orthopaedic Research Society , March 12-15 , 2000 , Orlando , Florida. 2000:48504.

24. The ROF. Connie Chen , Kwang Kim , B . S ., Savio L-Y . Woo , Ph . D ., D . Sc ., D . Eng . :1-2.

25. Arendt E a. MPFL reconstruction for PF instability. The soft (tissue) approach. *Orthop Traumatol Surg Res*. 2009;95(8):97-100. doi:10.1016/j.otsr.2009.09.002.

26. Victor J, Wong P, Witvrouw E, Sloten J Vander, Bellemans J. How isometric are the medial patellofemoral, superficial medial collateral, and lateral collateral ligaments of the knee? *Am J Sports Med*. 2009;37(10):2028-2036. doi:10.1177/0363546509337407.

27. Wismans J, Veldpaus F, Janssen J, Huson a, Struben P. A three-dimensional mathematical model of the knee-joint. *J Biomech*. 1980;13(8):677-685. doi:10.1016/0021-9290(80)90354-1.

28. Bollier M, Fulkerson J, Cosgarea A, Tanaka M. Technical failure of medial patellofemoral ligament reconstruction. *Arthroscopy*. 2011;27(8):1153-1159. doi:10.1016/j.arthro.2011.02.014.

29. Dixit S, DiFiori JP, Burton M, Mines B. Management of patellofemoral pain syndrome. *Am Fam Physician*. 2007;75(2):194-202.

30. Race a, Amis a a. The mechanical properties of the two bundles of the human posterior cruciate ligament. *J Biomech*. 1994;27(1):13-24. doi:0021-9290(94)90028-0 [pii].

31. Herbort M, Hoser C, Domnick C, et al. MPFL reconstruction using

a quadriceps tendon graft. Part 1: Biomechanical properties of quadriceps tendon MPFL reconstruction in comparison to the Intact MPFL. A human cadaveric study. *Knee*. 2014;21(6):1169-1174. doi:10.1016/j.knee.2014.07.026.

32. Squat exercise - should you prescribe the squat as a rehabilitation exercise? Here's a review of the evidence. http://www.sportsinjurybulletin.com/archive/squat-exercise.html#. Accessed April 13, 2016.

33. Malliaras P, Kamal B, Nowell A, et al. Patellar tendon adaptation in relation to load-intensity and contraction type. *J Biomech*. 2013;46(11):1893-1899. doi:10.1016/j.jbiomech.2013.04.022.

34. Ligaments of the Knee - Sports - Orthobullets.com. http://www.orthobullets.com/sports/3001/ligaments-of-the-knee. Accessed April 15, 2016.